INVESTMENTS

Ninth Canadian Edition · Bodie, Kane, Marcus

CHAPTER 11

The Efficient Market Hypothesis

Verified against source text — learning objectives, formulas, and worked examples checked directly against the chapter.

Study Guide · Prepared for Personal Use

LO1 · Random Walks and the Efficient Market Hypothesis

Early researchers (e.g., Maurice Kendall, 1953) were surprised to find **no predictable patterns** in stock prices — price changes appeared to wander randomly. Rather than a market flaw, this is actually the **signature of a well-functioning, informationally efficient market:** if any reliably predictable pattern existed, alert investors would trade on it immediately, competing away the profit opportunity and moving prices until the pattern disappeared. Predictable price patterns are, paradoxically, a sign of an inefficient market — truly efficient markets should produce prices that already reflect the value of any information, leaving only the *surprises* still to come as unpredictable, random future price changes. This is the **random walk** hypothesis.

Versions of the Efficient Market Hypothesis

EMH Form	What It Claims
Weak-form EMH	Prices reflect all information derivable from market trading data (past prices, trading volume, short interest). Implies trend/technical analysis based purely on price history is fruitless.
Semistrong-form EMH	Prices reflect all publicly available information, including fundamentals: product line, management quality, balance sheet, patents, earnings forecasts, accounting practices. Implies fundamental analysis based on public information alone cannot reliably generate abnormal returns.
Strong-form EMH	Prices reflect all information relevant to the firm, including information available only to company insiders. An extreme version — few dispute that corporate insiders can profit from privileged information before it becomes public, which is precisely why insider trading is regulated (e.g., Rule 10b-5 in the U.S.; provincial securities commissions in Canada).

> *Each stronger form of the EMH subsumes the weaker ones: if the strong form holds, the semistrong and weak forms must also hold (though not vice versa). Each successive form asserts efficiency with respect to a progressively broader information set.*

LO2 · Implications of the EMH

Technical Analysis

Technical analysis — searching for recurring, predictable patterns in past prices and volume — is directly contradicted by even the weak form of the EMH. If prices already reflect all information in past trading data, price histories provide no signal about future performance; any pattern that seemed to work would attract traders whose very activity of exploiting it would eliminate it.

Fundamental Analysis

Fundamental analysis — using publicly available economic and firm data to estimate intrinsic value — is largely undermined by the semistrong form. Since so many well-resourced analysts already pore over the same public data, it becomes very hard for any single analyst's effort to be the source of a durably exploitable insight, though skilled analysts may still be the ones whose research first moves prices to their fair value.

Active vs. Passive Portfolio Management

If markets are efficient, active management (trying to identify mispriced securities or time markets) is largely a wasted effort net of its costs, and a low-cost **passive strategy** — a well-diversified portfolio (often an index fund) with no attempt to beat the market — becomes highly attractive. Note the self-correcting paradox: if literally every investor believed markets were perfectly efficient and switched to pure indexing, no one would be left analyzing securities, and prices would drift away from fair value — creating profit opportunities for the few remaining active analysts. In practice, markets are probably efficient enough that active management struggles to beat its costs on average, but not so perfectly efficient that no analysis is worthwhile at all.

The Role of Portfolio Management Even Under Full Efficiency

Even if one fully accepts market efficiency, portfolio management still adds real value: appropriate diversification for an investor's specific circumstances, tailoring the risk profile to personal risk tolerance and horizon, and considering tax and liquidity needs specific to the individual investor — none of which "beating the market" addresses.

LO3 · Event Studies

The EMH gives rise to a powerful empirical research technique: since informationally efficient prices should move only in response to new information, an **event study** isolates the price impact of a specific event (e.g., a dividend change, an earnings announcement, a merger bid) by first estimating what the security's return *would have been* absent the event, then attributing the difference between actual and expected return to the event itself — the **abnormal return.**

The Market Model Approach

A common benchmark uses the single-index (market) model from Chapter 8: a security's return is modeled as a market-driven component (alpha plus beta times the market return) plus a firm-specific residual. The residual — the return left over after removing the part explained by overall market movements — is interpreted as the abnormal return attributable to firm-specific news, including the event under study (see Formulas page and worked Example on the Examples page).

A key methodological caution: the alpha and beta used as the benchmark must be estimated from a period **sufficiently separated in time** from the event itself, so the estimates are not contaminated by the very abnormal performance being studied. Because of this vulnerability, characteristic-matched portfolio benchmarks (matching on size, book-to-market, etc.) have become increasingly popular alternatives to the simple market model.

LO4 · Are Markets Efficient?

Decades of research have uncovered patterns that appear to challenge market efficiency — collectively known as **anomalies.** Whether these represent genuine mispricing or simply reflect a flawed risk-adjustment model (the **joint hypothesis problem:** every test of market efficiency is really a joint test of efficiency AND whatever asset-pricing model was used to define "normal" returns) remains actively debated.

Momentum and Reversal

Broad market indexes show only weak serial correlation, but individual stocks and sectors show more pronounced patterns: short-to-intermediate-horizon **momentum** (good/bad recent performance tends to continue for a period, documented in both U.S. and Canadian studies) alongside longer-horizon **reversal** (losers rebound and past winners fade back over multi-year horizons). One behavioural interpretation: short-run overreaction to news creates momentum, which then partially corrects into a longer-run reversal once the market recognizes its overreaction.

The Size (Small-Firm) Effect

Portfolios of small-capitalization stocks have shown consistently higher average annual returns than large-cap portfolios: sorting NYSE stocks into 10 size-based portfolios (1926–2015), the smallest-firm portfolio outperformed the largest-firm portfolio by **7.65% per year on average** — a gap that persists even after risk-adjusting with the CAPM. Related patterns include the **neglected-firm effect** (stocks with low analyst/institutional coverage earn a premium, especially pronounced in January) and the book-to-market effect (covered further in Chapter 13).

Post-Earnings-Announcement Drift

Stocks with positive earnings surprises continue to show **positive** abnormal returns for weeks to months *after* the announcement date — seemingly slow, predictable price adjustment to public information, in direct tension with the semistrong-form EMH.

Interpreting the Evidence — Behavioural vs. Risk-Based

Two broad camps interpret this evidence differently. **Behavioural finance** proponents (Chapter 12) argue these patterns reflect systematic irrationality in how investors process information. **Efficient-market defenders** argue the patterns may simply reflect a mismeasured risk premium (an incomplete asset-pricing model) or may not survive real-world implementation costs. A further practical limitation for genuine arbitrage: **short-selling costs and constraints**, the difficulty of borrowing shares to sell short, and the risk that mispricing can widen further before it corrects (i.e., you can be "right" but still lose money and be forced to close your position first) all impede a would-be arbitrageur from fully correcting overpriced securities — a limitation revisited in depth in Chapter 12.

LO5 · Mutual Fund and Analyst Performance

The most direct test of market efficiency: can skilled professionals actually generate consistent risk-adjusted abnormal profits after costs? The evidence on both stock analysts and mutual fund managers is genuinely mixed.

Stock Market Analysts

Analyst recommendations are historically skewed positive — in one large 1996 sample of 5,628 covered firms, the average recommendation was 2.04 on a 1 (strong buy) to 5 (strong sell) scale — so raw recommendation levels cannot be taken at face value. Studies instead focus on **changes** in recommendations: positive changes are associated with roughly **+5%** average price moves; negative changes with roughly **−11%** moves, and the impact appears permanent (consistent with analysts revealing genuine new information rather than just triggering temporary buy/sell pressure). However, portfolio strategies built on these recommendation changes would generate very heavy trading activity, and the associated costs plausibly wipe out much of the apparent profit. Canadian studies (Dutta et al.; Brown, Richardson & Trzcinka) find broadly similar patterns, including sensitivity of abnormal-return duration to the direction of earnings revisions (persisting roughly 9 months for upward revisions versus 3 months for downward revisions in one study).

Mutual Fund Managers

As previewed in Chapter 4, the broad pattern across decades of research is that the **average** actively managed equity mutual fund has *not* outperformed a passive market-index benchmark on a risk-adjusted, net-of-fees basis. Some funds do show a period of outperformance, but persistence of that outperformance from one period to the next is weak and inconsistent — making it very difficult for investors to identify skilled managers in advance rather than merely with hindsight.

Chapter 11 — Big-Picture Takeaways

- Unpredictable ("random walk") price changes are the expected signature of an efficient market, not evidence of dysfunction.
- The three EMH forms (weak, semistrong, strong) differ by their information set; each stronger form subsumes the weaker ones.
- Event studies use a benchmark model (often the single-index market model) to isolate abnormal returns attributable to a specific news event.
- Documented anomalies (momentum/reversal, the size effect, post-earnings drift) genuinely challenge simple market efficiency, but interpretation is contested due to the joint-hypothesis problem and practical limits to arbitrage.
- Analyst recommendation changes carry real, permanent price impact, but trading on them incurs costs that likely offset much of the apparent profit.

- The average actively managed mutual fund has historically underperformed passive benchmarks net of fees, and outperformance shows weak persistence across time — the strongest practical evidence for market efficiency.

The event-study abnormal-return calculation (F1–F2) is the core quantitative skill from this chapter.

F1 | Market Model (Single-Index Benchmark)

$$r_t = \alpha + \beta \times r_{Mt} + e_t$$

r_t = security's return in period t; r_{Mt} = market return in period t; α, β estimated from a period well-separated from the event. e_t is interpreted as the firm-specific (abnormal) return.

F2 | Abnormal Return (Event-Period Residual)

$$e_t = r_t - (\alpha + \beta \times r_{Mt})$$

The abnormal return is the security's actual return minus its expected return given that period's actual market return — the portion of return attributable to the firm-specific event under study.

F3 | Size (Small-Firm) Effect — Historical Spread

$$\text{Return Spread} = \text{Return(smallest-firm decile)} - \text{Return(largest-firm decile)} = 7.65\%/\text{yr}\ (1926\text{–}2015)$$

A widely cited empirical anomaly benchmark: small-cap portfolios have historically outperformed large-cap portfolios by this margin annually, even after CAPM risk adjustment.

EXAMPLE 11.3 | Computing an Abnormal Return

Given: An analyst has estimated α = 0.05% and β = 0.8 for a stock. On a day the market rises 1%, the stock actually rises 2%.

Step 1 — Expected (predicted) return given market move:

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Expected = 0.05% + 0.8 × 1% = 0.85%
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Step 2 — Abnormal return:

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Abnormal return = 2% − 0.85% = 1.15%
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The analyst infers that firm-specific news that day caused an additional 1.15% return beyond what broad market movements alone would predict — this 1.15% is the estimated impact of whatever event occurred.

EXAMPLE | The Size Effect in Practice

Given: NYSE stocks sorted annually (1926–2015) into 10 portfolios by total equity market value.

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Return(Portfolio 1, smallest firms) − Return(Portfolio 10, largest firms) = 7.65% per year
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This spread persists even after adjusting for CAPM beta — the smaller portfolios are riskier, but not enough (by CAPM logic) to fully explain a gap this large. Note the practical puzzle the chapter raises: firm size is nearly costless to observe, so if this premium were reliably exploitable, it seems surprising that competition among investors has not eliminated it — a tension at the heart of the anomalies literature.

PRACTICE PROBLEM | Elbow Falls Research Group

Scenario

You are a junior analyst at Elbow Falls Research Group in Calgary, studying the market impact of a surprise CEO resignation announcement for a TSX-listed company, PrairieCore Industries (α = 0.10%, β = 1.1, estimated from a clean prior period).

Part A — EMH Forms

(i) If PrairieCore's stock price reacts immediately and fully to the resignation announcement on the day it is made public, and shows no further drift afterward, which form(s) of the EMH is this behaviour consistent with?

(ii) If instead insiders had already traded profitably on advance knowledge of the resignation before the public announcement, which form of the EMH would this violate?

Part B — Event Study Calculation

On the announcement day, the S&P/TSX Composite rises 0.4%. PrairieCore's stock actually falls 3%.

(i) Calculate PrairieCore's expected return for the day using the market model.

(ii) Calculate the abnormal return attributable to the resignation announcement.

(iii) Explain in one sentence why Elbow Falls should estimate α and β from a period well before the announcement, rather than including recent data.

Part C — Anomalies and the Joint Hypothesis Problem

A colleague claims PrairieCore (a small-cap stock) should be added to a "small-firm effect" trading strategy, citing the 7.65%/year historical size premium.

(i) Explain the joint hypothesis problem as it applies to this claim — what two things is a test of the size effect actually testing at once?

(ii) Name one behavioural and one risk-based explanation that could each be offered for why small firms have historically earned higher average returns.

Part D — Limits to Arbitrage

Suppose Elbow Falls' analysis suggests PrairieCore is now undervalued after its overreaction to the resignation news, and recommends shorting a comparable overvalued peer while going long PrairieCore.

(i) Identify two practical impediments (from LO4) that could prevent this strategy from working as cleanly as the theory suggests.

(ii) Explain how post-earnings-announcement drift, if it applied here, might create a profit opportunity — and why that opportunity would still be difficult for Elbow Falls to fully exploit given real-world trading costs.

Hints & Guidance

Part A tests LO1. Part B tests LO3 and Formulas F1–F2 (mirror Example 11.3 exactly). Part C tests LO4 and the joint hypothesis problem specifically. Part D synthesizes LO4's limits-to-arbitrage discussion with the anomalies covered.